

The biochemical alterations following administration of Kalpaamruthaa and *Semecarpus anacardium* in mammary carcinoma

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Abstract

Background: Breast cancer is one of the most common cancers in women of developed and developing countries. Lipids, lipoproteins and lipid-metabolizing enzymes have been associated with the risk of breast cancer. Kalpaamruthaa (KA) is a modified Siddha preparation, which contains *Semecarpus anacardium* Linn. (SA), *Emblica officinalis* (EO) and honey.

Objective: The present study was embarked to study the variations in lipids, lipid-metabolizing enzymes and lipoproteins in cancerous animals and the effect of KA on the lipid metabolism.

Materials and methods: Breast cancer was induced in rats by administering 7,12-dimethylbenz(a)anthracene orally (25 mg/kg body weight). After 90 days of induction, KA (300 mg/kg body weight) and SA (200 mg/kg body weight) were administered for 14 days, by gastric intubations. The levels of lipids and lipid-metabolizing enzymes were analysed in control and experimental animals.

Results and conclusion: The increased levels of total cholesterol, free cholesterol, phospholipids, triglycerides and free fatty acids and decreased levels of ester cholesterol in plasma, liver and kidney found in cancer suffering animals were reverted back to near normal levels on treatment with KA and SA. In mammary carcinoma bearing animals, the activities of total lipase, cholesterol ester synthase, and cholesterol ester hydrolase were significantly ($p < 0.05$) increased whereas lipoprotein lipase and lecithin cholesterol-acyl transferase were decreased. The levels of very low-density lipoprotein (VLDL) and low-density lipoprotein (LDL) were increased and the level of high-density lipoprotein (HDL) was decreased. These alterations were recouped back upon treatment with KA as well as SA when compared to cancer animals. The effects of KA were found to be more effective than SA. No significant alterations were observed in herbal preparation control animals when compared to control animals.

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1. Introduction

Worldwide, breast cancer is the most common cancer among women after skin cancer, and it is also the second leading cause of cancer death (after lung cancer) in women [1]. The mechanisms of action through

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which the lipids participate in the growth of breast cancer are beginning to be known. Breast tumour cells require lipids as a metabolic fuel. In this respect it has been described that fatty acid synthase, a major enzyme of fatty acid biosynthesis, is produced in primary breast tumours and certain human and murine cancer cell lines. In human tumours its over expression is associated with poor prognosis [2]. Furthermore, genes whose function is to activate final genes of fatty acid synthesis have been found both amplified and expressed in several breast cancer-derived cell lines [3]. Conversely, the pharmacological inhibition of fatty acid synthesis in breast cancer cells with fatty acid synthase over expression caused reduced growth and programmed cell death, an effect that was reversed by addition of fatty acids to the growth media [4].

Peroxidation of lipids in biomembranes and tissues causes the leaking of these lipids into circulation, and consequently leads to hyperlipidemia. The direct relationships between lipid peroxidation and subsequent alteration in lipid metabolism have already been well established [5].

Both benign and malignant proliferation of breast tissue in women has been associated with changes in plasma lipids and lipoprotein levels. Lane et al. [6] reported that prior to diagnostic biopsy, serum lipid and apolipoprotein components of low-density lipoproteins were increased in women with fibrocystic disease and early stage breast cancer but decreased in women with early recurrence.

Various compounds characterized from natural antioxidants, such as vitamins and flavonoids, have been tested for their potential to cure diseases by reducing oxidative stresses. Kalpaamruthaa (KA) is a modified Siddha preparation, which has been formulated in our laboratory. It contains *Semecarpus anacardium* Linn. (SA), *Emblica officinalis* (EO) and honey in different ratio. The components studies of this herbal preparation reveal the presence of flavonoids, ascorbic acid, polyphenols, tannins, sugars, sterols, etc. Dose-dependent study of Kalpaamruthaa was carried out in mammary carcinoma bearing rats, which showed fixed 300 mg/kg body weight as the effective dosage levels [7].

S. anacardium Linn. (Family: *Anacardiaceae*) (SA) is distributed in Sub-Himalayan region, tropical and central parts of India. The nut is commonly called 'marking nut' and in the vernacular as 'Ballataka' or 'Bhilawa' [8]. Due to its wide therapeutic utility and easy availability, the chemistry of *S. anacardium* nut extract was extensively studied. Many pharmacological properties such as antirheumatic [9] and anticancer [10,11] have been

attributed to the nuts of the plant. Phytochemical studies of Siddha preparation of *S. anacardium* nut extract from our laboratory, reported by [12] was found to contain phenols, flavonoids and carbohydrates.

E. officinalis (syn. *Phyllanthus emblica* L.) Eupherbiaceae is used therapeutically in Indian system of medicine. Fruits of EO are used for the treatment of a number of diseases like antibacterial [13], anti-inflammatory agent [14] and modify metal-induced clastogenic effects [15].

Researchers now believe that ascorbic acid, a major component of EO prevents cancer [16]. Extensive animal, clinical and epidemiological studies were carried out on the role of ascorbic acid in the prevention of different types of cancers. Vitamin C has an ability to convert cancerous cells to normal ones. There is evidence that vitamin C has beneficial effects on lipid-metabolizing enzymes, as it promotes the activities of lipoprotein lipase [17] and implications are that, Vitamin C inhibits abnormalities.

Flavonoids are members of large chemical groups, polyphenols, which have known preventive activities against degenerative diseases, e.g. cancer and coronary heart disease [18]. Flavonoids are biosynthesized by plants and have strong antioxidant activity such as scavenging free radicals that are involved in cell damage and tumour promotion [19]. The current investigation has been carried out to study the effects of KA and SA on the alterations in the lipid metabolism caused by mammary carcinoma.

2. Materials and methods

2.1. Animals and diet

Female albino Sprague–Dawley rats of Wistar stain, 50–55 days, were obtained from the National Institute of Nutrition, Hyderabad. The animals were housed in spacious hygienic plastic cages in an air-conditioned room controlled for temperature ($27 \pm 1^\circ\text{C}$), humidity ($60 \pm 5\%$) and lighting 12h: light–dark cycle) and they were given food (Gold Mohur rat feed, Hindustan lever Limited, Mumbai, India) and water ad libitum.

2.2. Chemicals

7,12-Dimethylbenz(a)anthracene (DMBA), was obtained from Sigma Chemical Company, St. Louis, USA. Ammonium molybdate, silicic acid, sodium metaperiodate, *p*-nitrophenyl acetate were obtained from Sisco Research Laboratories Pvt. Ltd., Mumbai, India.

2.3. Animal model

The rats were divided into five groups with six animals in each group and were given the dose regimen as follows. Group I: control animals, Group II: breast cancer was induced in overnight-fasted animals by a single dose of DMBA in olive oil (25 mg/kg body weight) [20,21] by gastric intubation. After 6 weeks, adenocarcinoma was confirmed by histologic examination. Group III: breast cancer-induced animals (as in Group II) were treated with the herbal preparation, Kalpaamruthaa (300 mg/kg body weight/day) in olive oil orally by gastric intubation for 14 days. Group IV: breast cancer-induced animals (as in Group II) were treated with the herbal preparation, *S. anacardium* (200 mg/kg body weight/day) in olive oil orally by gastric intubation for 14 days. Group V: Kalpaamruthaa-treated control, the herbal preparation (300 mg/kg body wt) was administered to control animals for 14 days by gastric intubation.

After the experimental period (3 months after induction + 14 days treatment), the animals were sacrificed by decapitation. Liver and kidney were removed immediately, washed well with ice-cold saline and homogenized in Tris–HCl buffer (0.1 M, pH 7.4). Blood was also collected for further analyses.

2.3.1. Extraction of lipids

The lipids were extracted and quantified by the method of Folch et al. [22]. A known volume of suspension was mixed with chloroform–methanol mixture and homogenate was filtered through Whatmann filter paper (No. 1) into a separating funnel. The residue of the filtration was scraped and homogenized in chloroform. The residue was once again scraped from the filter paper and homogenized with chloroform–methanol mixture (2:1, v/v) and the resulting filtrate was evaporated to dryness. The weight of the dried lipid was taken as the total lipid content. The delipidated residue was scraped from the filter paper was used for further analysis.

2.3.2. Folch wash

The dried lipid residue after evaporation was redissolved in chloroform–methanol and mixed with 0.1 M KCl. The contents were shaken well and the upper aqueous phase containing ganglioside and other water-soluble compounds were separated. The lower phase contains neutral and phospholipids in chloroform were again washed with Folch's reagent (0.1N, CH₃OH:CHCl₃) and the upper aqueous phase was aspirated. The lower chloroform phase was made up to a

known volume and aliquots were taken for further analysis.

2.4. Assay of lipids

2.4.1. Total cholesterol

Total cholesterol (TC) was estimated by the method of Parekh and Jung [23]. Tissue lip extract from the chloroform phase (after Folch wash) was taken. From this ferric chloride–uranyl acetate reagent and sulphuric acid–ferrous sulphate reagent was added. After 20 min, the colour developed was read at 540 nm using a Photochem colorimeter.

2.4.2. Free cholesterol (FC)

Free cholesterol (FC) was measured by the method of Leffler and McDougald [24]. The difference between total cholesterol and free cholesterol is the ester cholesterol (EC) content.

2.4.3. Phospholipids (PL)

Phospholipid was estimated by the method of Rouser et al. [25]. Lipid extract was diluted with distilled water. To this perchloric acid (70%) was added and digested on a sand bath till the solution became colourless ammonium molybdate (3%) and ascorbic acid (3%) were added and the mixture was kept in a boiling water bath for 6 min. The blue colour developed was read at 710 nm using a Photochem colorimeter.

2.4.4. Triglyceride (TG)

Triglyceride (TG) in plasma was estimated by the method of Rice [26] based on the method of Van Handel [27]. To Folch wash aliquot, chloroform–methanol mixture (2:1, v/v) was added. Activated silicic acid was added to it, shaken and allowed to stand to 30 min. After centrifugation, supernatant was taken. From that alcoholic KOH solution (0.4%) was added and the mixture was saponified in a 60–70 °C water bath for 20 min. To this, 0.2N H₂SO₄ was added and kept in boiling water bath for 10 min. After cooling the tubes, sodium metaperiodate (0.1 M) was added stand it for 10 min and chromotropic acid was added kept in boiling water bath for 30 min. After cooling thiourea solution (7%) was added and the colour developed was read at 570 nm using a Photochem colorimeter.

2.4.5. Free fatty acid

Free fatty acids were estimated by the method of Horn and Menahan [28]. Folch wash aliquot was mixed with chloroform–heptane–methanol (200:150:7, v/v) solvent and was shake well. To this activated silicic acid was

added shaken and left aside for 30 min. The solution was then centrifuged and the supernatant was transferred to tubes containing Cu-TEA solution. The content were agitated, the mixture was separated into two phases by centrifugation. Upper phase was transferred to another tube and mixed with diethyl dithiocarbamate (0.1%) solution. The yellow colour developed was read at 430 nm using a Photochem colorimeter.

2.5. Assay of lipid-metabolizing enzymes

2.5.1. Total lipase

Plasma lipid-metabolizing enzymes total lipase was measured by the method of Bier [29]. In tissue homogenate, buffer (phosphate buffer 0.1 M pH 7.2) and distilled water were added and incubated at 25 °C for 5 min. Then, substrate (*p*-nitrophenyl acetate was dissolved in methanol) was added. The colour developed was read at 400 nm after 20 min along with standards in a Photochem colorimeter.

2.5.2. Lipoprotein lipase

Lipoprotein lipase was measured by the method of Schmidt [30]. In olive oil emulsion, deoxycholate (10 mM), buffer (triethanolamine buffer 1.0 M pH 8.5) and tissue homogenate were incubated at 37 °C for 30 min keeping the tubes in a boiling water bath for 1 min arrested the reaction. To this chloroform and doles mixture (isopropanol, hexane and 2N sulphuric acid were mixed in the ratio of 40:10:1, v/v) were added and mixed well. Then, hexane and distilled water were added, shaken well and centrifuged. Supernatant was taken and 0.01N sulphuric acids was added and centrifuged, from that supernatant, chloroform and copper reagent were added. From this, aliquot was taken and DETC (diethyl dithiocarbamate 0.1%) reagent was added. The colour developed was read at 420 nm in a Photochem colorimeter.

2.5.3. Lecithin cholesterol-acyl transeferase (LCAT)

The activity of the enzyme was assayed by the method of Legraud et al. [31] with modifications of [32]. The incubation mixture contained the substrate (dextran sulphate 0.2%) and tissue homogenate, the mixture was immediately transferred to a tube containing isopropanol to arrest the reaction. This gave the cholesterol present at the beginning of the experiment. Incubation at 27 °C was continued for 90 min when another 0.4 ml was pipetted out and the reaction was arrested. Incubation was further continued for the next 90 min and the reaction was arrested. After centrifugation, to the supernatant, acetone and digitonin were added. It was left aside for 1 h, cen-

trifuged and processed for the estimation of cholesterol by the method of Parkeh and Jung [23].

2.5.4. Cholesterol ester hydrolase (CEH)

Cholesterol ester hydrolase activity was assayed by the method of Kothari et al. [33] with slight modification by Krichevsky and Kothari [34]. Acetate buffer (0.1 M pH 4.0), substrate and tissue homogenate were mixed well and incubated at 37 °C for 6 h with occasional shaking. The reaction was stopped by the addition acetone–ethanol mixture and centrifuged. Digitonin was added to the supernatant followed by drops of acetic acid. The tubes were closed and kept in a dark chamber for 16 h, and centrifuged. The precipitate was washed twice with acetone–ether mixture and then with dry ether. Ether was evaporated off and the tubes were heated in a sand bath at 110 °C for 30 min. The precipitate was then dissolved in FUA reagent and cholesterol content was estimated by the method of Parkeh and Jung [23].

2.5.5. Cholesterol ester synthetase (CES)

Cholesterol ester synthetase was assayed by the method of Kothari et al. [33]. The incubation mixture containing acetate buffer (0.1 M pH 4.0), substrate and tissue homogenate was incubated at 37 °C for 6 h with occasional shaking. The reaction was arrested by the addition of acetone–ethanol mixture and centrifuged. Digitonin was added to the supernatant, followed by drops of acetic acid. The contents were closed and kept in a dark chamber for 16 h. The precipitate, after centrifugation, was washed twice with acetone–ether mixture and finally with dry ether. The cholesterol content was estimated by the method of Parkeh and Jung [23].

2.6. Assay of lipoproteins

Lipoproteins were fractionated by a dual precipitation technique of Wilson and Spiger [35]. Addition of heparin–magnesium to plasma, which precipitated very low-, low-, and high-density lipoproteins were left in the supernatant and the content was measured in the fraction.

2.6.1. HDL

Plasma was added to heparin–magnesium reagent and mixed well. The solution was allowed to stand at 4 °C for 30 min and centrifuged at 2000 × *g* and maintained at 10 °C for 30 min. The supernatant contained the high-density lipoprotein fraction.

2.6.2. Low- and very low-density lipoproteins

Plasma was added to 10% SDS. The content were mixed well and incubated at 37 °C for 2 h. The con-

Table 1
Effect of KA on the levels of lipids in plasma of control and experimental animals

Parameters (mg/dl)	Group I (control)	Group II (tumour induced)	Group III (tumour + KA treated)	Group IV (tumour + SA treated)	Group V (control + KA)	p-Value
Total cholesterol	141.31 ± 13.6	186.58 ± 18.1 ^{a,*}	155.40 ± 14.9 ^{b,*}	164.50 ± 15.8 ^{b,c,*}	140.13 ± 13.8 ^{a,NS}	<0.05
Free cholesterol	86.20 ± 8.1	150.07 ± 14.4 ^{a,*}	121.27 ± 11.6 ^{b,*}	98.27 ± 9.3 ^{b,c,*}	85.73 ± 7.9 ^{a,NS}	<0.05
Ester cholesterol	55.11 ± 4.9	36.51 ± 3.2 ^{a,*}	52.74 ± 4.7 ^{b,*}	48.12 ± 4.1 ^{b,c,*}	54.40 ± 4.9 ^{a,NS}	<0.05
Phospholipid	110.73 ± 10.4	171.28 ± 16.6 ^{a,*}	127.21 ± 12.1 ^{b,*}	139.10 ± 13.2 ^{b,c,*}	108.95 ± 10.3 ^{a,NS}	<0.05
Triglycerides	42.87 ± 3.7	74.05 ± 6.8 ^{a,*}	58.31 ± 5.2 ^{b,*}	49.01 ± 4.3 ^{b,c,*}	41.62 ± 3.6 ^{a,NS}	<0.05
Free fatty acid	21.27 ± 0.17	38.41 ± 3.2 ^{a,*}	25.29 ± 1.9 ^{b,*}	27.21 ± 2.2 ^{b,c,*}	21.27 ± 0.16 ^{a,NS}	<0.05

Values are expressed as mean ± S.D. for six animals. NS: non-significant.

^a Denotes when compared with Group I.

^b Denotes when compared with Group II.

^c Denotes when compared with Group III.

* Denotes $p < 0.05$.

tent were centrifuged at $1000 \times g$ for 15 min. Very low-density lipoprotein aggregated as a pellicle at the top. The supernatants contained the high- and low-density lipoprotein fractions. Plasma lipoprotein cholesterol content is expressed as mg/dl:

1. VLDL cholesterol = total cholesterol–subnatant from SDS.
2. LDL cholesterol = subnatant from SDS–heparin–magnesium chloride supernatant.

2.7. Statistical analysis

Results were presented as mean ± S.D. of six rats. The results were statistically evaluated using student's *t*-test and ANOVA and post hoc-Newman-Keuls test.

3. Results

3.1. Lipids

The levels of lipids (total cholesterol, free cholesterol, ester cholesterol, Phospholipids, triglycerides and free fatty acids) in plasma, liver and kidney of control and experimental animals were depicted in Tables 1–3, respectively. The levels of total cholesterol, free cholesterol, phospholipids, triglycerides and free fatty acids in plasma and liver were increased ($p < 0.05$) significantly whereas the level of EC was decreased significantly ($p < 0.05$) in cancer suffering animals when compared to control animals. Upon herbal preparation administration (SA and KA), levels were significantly reverted back to the near normal levels when compared to untreated cancer animals. The effect of KA was found to be more significant ($p < 0.05$) than in animals treated with SA.

No significant changes were observed in drug control animals when compared to control animals.

3.2. Lipid-metabolizing enzymes

A critical analysis of Tables 4 and 5 and Fig. 1 explains the alterations in the levels of lipid-metabolizing enzymes in liver, kidney and plasma, respectively, in control and experimental animals. The levels of total lipase, cholesterol ester synthetase, and cholesterol ester hydrolase were significantly ($p < 0.05$) increased in cancer bearing animals when compared to control animals. On the other hand lipoprotein lipase and lecithin cholesterol-acyl transferase, were decreased significantly ($p < 0.05$) in liver and kidney of cancer-induced animals when compared to control animals. Administration of herbal preparations (SA and KA) significantly ($p < 0.05$) brought

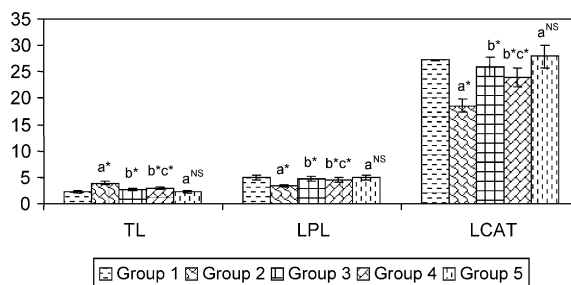


Fig. 1. Effect of KA on lipid-metabolizing enzymes in plasma of control and experimental animals. Values are expressed as mean ± S.D. for six animals. Comparisons are made between: (a) Groups I and II; (b) Groups II and III; (b*) Groups II and IV; (c) when compared with Groups III and NS–Groups V and I. Statistical significance are expressed as: $p < 0.05$; and NS: non-significant. Units — total lipase: μmol of *p*-nitrophenol liberated/h/mg protein; lipoprotein lipase: μmol of free fatty acid liberated/h/mg protein; lecithin cholesterol-acyl transferase: μmol of cholesterol esterified/h/mg protein.

Table 2

Effect of KA on lipids in liver of control and experimental animals

Parameters (mg/dl)	Group I (control)	Group II (tumour induced)	Group III (tumour + KA treated)	Group IV (tumour + SA treated)	Group V (control + KA)	p-Value
Total cholesterol	3.58 ± 0.31	7.94 ± 0.72 ^{a,*}	4.35 ± 3.71 ^{b,*}	5.02 ± 4.6 ^{b,c,*}	3.57 ± 3.0 ^{a,NS}	<0.05
Free cholesterol	1.97 ± 0.15	4.78 ± 0.44 ^{a,*}	2.04 ± 1.66 ^{b,*}	2.89 ± 2.3 ^{b,c,*}	1.96 ± 1.5 ^{a,NS}	<0.05
Ester cholesterol	1.61 ± 0.12	3.16 ± 0.27 ^{a,*}	1.98 ± 1.54 ^{b,*}	2.31 ± 1.9 ^{b,c,*}	1.51 ± 1.1 ^{a,NS}	<0.05
Phospholipid	6.84 ± 0.63	9.91 ± 0.95 ^{a,*}	7.05 ± 6.61 ^{b,*}	7.86 ± 7.31 ^{b,c,*}	6.82 ± 6.53 ^{a,NS}	<0.05
Triglycerides	9.16 ± 0.87	13.83 ± 1.09 ^{a,*}	9.96 ± 9.42 ^{b,*}	10.11 ± 0.07 ^{b,c,*}	9.15 ± 0.86 ^{a,NS}	<0.05
Free fatty acid	6.02 ± 0.56	9.48 ± 0.89 ^{a,*}	6.85 ± 0.62 ^{b,*}	7.15 ± 0.67 ^{b,c,*}	6.02 ± 0.56 ^{a,NS}	<0.05

Values are expressed as mean ± S.D. for six animals. NS: non-significant.

^a Denotes when compared with Group I.^b Denotes when compared with Group II.^c Denotes when compared with Group III.* Denotes $p < 0.05$.

Table 3

Effect of KA on lipids in kidney of control and experimental animals

Parameters (mg/dl)	Group I (control)	Group II (tumour induced)	Group III (tumour + KA treated)	Group IV (tumour + SA treated)	Group V (control + KA)	p-Value
Total cholesterol	2.15 ± 0.16	5.65 ± 0.51 ^{a,*}	2.53 ± 0.21 ^{b,*}	2.81 ± 0.23 ^{b,c,*}	2.14 ± 0.17 ^{a,NS}	<0.05
Free cholesterol	1.27 ± 0.07	3.62 ± 0.31 ^{a,*}	1.81 ± 0.14 ^{b,*}	2.15 ± 0.17 ^{b,c,*}	1.27 ± 0.22 ^{a,NS}	<0.05
Ester cholesterol	0.88 ± 0.03	2.03 ± 0.17 ^{a,*}	1.09 ± 0.06 ^{b,*}	1.41 ± 0.09 ^{b,c,*}	0.87 ± 0.03 ^{a,NS}	<0.05
Phospholipid	6.21 ± 0.59	8.91 ± 0.84 ^{a,*}	7.08 ± 0.66 ^{b,*}	7.35 ± 0.69 ^{b,c,*}	6.21 ± 0.56 ^{a,NS}	<0.05
Triglycerides	6.95 ± 0.65	11.53 ± 0.7 ^{a,*}	7.32 ± 6.8 ^{b,*}	7.92 ± 0.75 ^{b,c,*}	6.85 ± 0.63 ^{a,NS}	<0.05
Free fatty acid	4.13 ± 0.36	6.85 ± 0.63 ^{a,*}	4.47 ± 0.39 ^{b,*}	4.95 ± 0.43 ^{b,c,*}	4.13 ± 0.36 ^{a,NS}	<0.05

Values are expressed as mean ± S.D. for six animals. NS: non-significant.

^a Denotes when compared with Group I.^b Denotes when compared with Group II.^c Denotes when compared with Group III.* Denotes $p < 0.05$.

Table 4

Effect of KA on lipid-metabolizing enzymes in liver of control and experimental animals

Parameters	Group I (control)	Group II (tumour induced)	Group III (tumour + KA treated)	Group IV (tumour + SA treated)	Group V (control + KA)	p-Value
Total lipase (μmol of <i>p</i> -nitrophenol liberated/h/mg protein)	5.62 ± 0.51	7.31 ± 0.67 ^{a,*}	5.89 ± 0.52 ^{b,*}	6.08 ± 0.53 ^{b,c,*}	5.60 ± 0.51 ^{a,NS}	<0.05
Lipoprotein lipase (μmol of FFA liberated/h/mg protein)	7.43 ± 0.67	2.59 ± 1.9 ^{a,*}	7.01 ± 0.62 ^{b,*}	5.98 ± 0.52 ^{b,c,*}	7.44 ± 0.68 ^{a,NS}	<0.05
Lecithin cholesterol-acyl transferase (μmol of cholesterol esterified/h/mg protein)	33.95 ± 2.9	17.31 ± 1.2 ^{a,*}	29.99 ± 2.3 ^{b,*}	27.21 ± 2.1 ^{b,c,*}	33.95 ± 2.7 ^{a,NS}	<0.05
Cholesterol ester synthetase (μmol of cholesterol esterified/h/mg protein)	21.07 ± 0.16	29.05 ± 2.3 ^{a,*}	23.91 ± 1.8 ^{b,*}	24.50 ± 1.8 ^{b,c,*}	21.04 ± 0.16 ^{a,NS}	<0.05
Cholesterol ester hydrolase (μmol of cholesterol liberated/h/mg protein)	7.23 ± 0.66	14.31 ± 1.8 ^{a,*}	9.09 ± 0.84 ^{b,*}	11.19 ± 0.06 ^{b,c,*}	7.23 ± 0.65 ^{a,NS}	<0.05

Values are expressed as mean ± S.D. for six animals. NS: non-significant.

^a Denotes when compared with Group I.^b Denotes when compared with Group II.^c Denotes when compared with Group III.* Denotes $p < 0.05$.

Table 5
Effect of KA on lipid-metabolizing enzymes in kidney of control and experimental animals

PARAMETERS	Group I (Control)	Group II (tumour induced)	Group III (tumour + KA treated)	Group IV (tumour + SA treated)	Group V (control + KA)	<i>p</i> -Value
Total lipase (μmol of <i>p</i> -nitrophenol liberated/h/mg protein)	3.94 $\pm$ 0.32	5.12 $\pm$ 0.44 ^{a,*}	4.32 $\pm$ 0.37 ^{b,*}	4.59 $\pm$ 0.39 ^{b,c,*}	3.93 $\pm$ 0.35 ^{a,NS}	<0.05
Lipoprotein lipase (μmol of FFA liberated/h/mg protein)	0.79 $\pm$ 0.01	0.38 $\pm$ 0.019 ^{a,*}	0.66 $\pm$ 0.04 ^{b,*}	0.53 $\pm$ 0.08 ^{b,c,*}	0.79 $\pm$ 0.02 ^{a,NS}	<0.05
Lecithin cholesterol-acyl transferase (μmol of cholesterol esterified/h/mg protein)	5.01 $\pm$ 0.44	4.19 $\pm$ 3.6 ^{a,*}	4.85 $\pm$ 0.42 ^{b,*}	4.76 $\pm$ 0.41 ^{b,c,*}	5.02 $\pm$ 0.44 ^{a,NS}	<0.05
Cholesterol ester synthetase (μmol of cholesterol esterified/h/mg protein)	14.27 $\pm$ 0.09	19.21 $\pm$ 1.2 ^{a,*}	15.65 $\pm$ 1.1 ^{b,*}	16.27 $\pm$ 1.2 ^{b,c,*}	14.21 $\pm$ 0.08 ^{a,NS}	<0.05
Cholesterol ester hydrolase (μmol of cholesterol liberated/h/mg protein)	2.40 $\pm$ 0.17	3.94 $\pm$ 0.34 ^{a,*}	2.75 $\pm$ 2.2 ^{b,*}	3.07 $\pm$ 0.24 ^{b,c,*}	2.38 $\pm$ 0.16 ^{a,NS}	<0.05

Values are expressed as mean $\pm$ S.D. for six animals. NS: non-significant.

^a Denotes when compared with Group I.

^b Denotes when compared with Group II.

^c Denotes when compared with Group III.

* Denotes $p < 0.05$.

Table 6
Effect of KA on lipoprotein levels in plasma of control and experimental animals

Parameters (mg/g wet tissue)	Group I (control)	Group II (tumour induced)	Group III (tumour + KA treated)	Group IV (tumour + SA treated)	Group V (control + KA)	<i>p</i> -Value
HDL cholesterol	18.40 $\pm$ 0.09	11.2 $\pm$ 0.05 ^{a,*}	16.7 $\pm$ 0.07 ^{b,*}	15.8 $\pm$ 0.07 ^{b,c,*}	18.5 $\pm$ 0.08 ^{a,NS}	<0.05
VLDL cholesterol	16.90 $\pm$ 0.07	19.2 $\pm$ 0.09 ^{a,*}	17.4 $\pm$ 0.07 ^{b,*}	18.1 $\pm$ 0.08 ^{b,c,*}	16.7 $\pm$ 0.05 ^{a,NS}	<0.05
LDL cholesterol	26.4 $\pm$ 0.10	34.9 $\pm$ 0.15 ^{a,*}	28.1 $\pm$ 0.12 ^{b,*}	30.5 $\pm$ 0.15 ^{b,c,*}	26.2 $\pm$ 0.11 ^{a,NS}	<0.05

Values are expressed as mean $\pm$ S.D. for six animals. NS: non-significant.

^a Denotes when compared with Group I.

^b Denotes when compared with Group II.

^c Denotes when compared with Group III.

* Denotes $p < 0.05$.

back all the altered enzyme levels to near normal levels when compared to mammary carcinoma bearing animals. KA was found to be more effective than SA. No significant alterations were observed in drug control animals when compared to control animals.

3.3. Lipoproteins

The plasma of mammary carcinoma bearing animals showed a profound increment in the levels of very low-density lipoprotein cholesterol (VLDL) and low-density lipoprotein cholesterol (LDL) and decreased high-density lipoprotein cholesterol (HDL) (Table 6). KA more significantly reverted back these altered lipoprotein levels when compared to SA. No significant changes were observed in all these parameters in drug control animals when compared to control animals.

4. Discussion

A number of studies have indicated the correlation of lipids and lipoproteins with the risk of breast cancer. The exact mechanism by which lipids, lipoproteins contribute to carcinogenesis are not clearly understood. An earlier study reported that lipids might primarily affect the gonads and subsequently higher estradiol secretion could influence the development of malignancies in the mammary glands and lymphoid system [36].

In the present study, significantly increased plasma total cholesterol and triglycerides levels were observed among the breast cancer animals. Our results also hypothesize that higher concentration of these may either play a role in carcinogenesis or are responsible for higher incidence of breast cancer. There are possible mechanisms through which lipoproteins contribute to carcinogenesis. The progression and proliferation of the tumour

cells depend on the elevation of cholesterol and phospholipid levels that modifies the lipid fluidity of the tumour cell membranes, thereby increasing the rate of malignancy in tumorous conditions [37].

The major metabolic vital organ, the liver plays a key role in cholesterol metabolism, in mammals. Reports reveal that, in animals, altered levels and changes in distribution of total cholesterol were observed during progression of tumour growth [38]. Kumar et al. [39] also have reported that breast cancer patients have higher lipid levels when compared to their respective control subjects. Elevation in the plasma and membrane cholesterol increases the fluidity of tumour membrane [40], which relaxes the contact inhibition of tumour cells and hence, invades to any part of the body, where it induces secondary tumours. The abnormal levels of lipid-metabolizing enzymes are proportionate or dependent on the stage of the malignancy. Lipids and their metabolizing enzymes affect the multistage carcinogenesis process, adding gravity to a causal effect.

Hyperlipidemia is an associated event that is secondary to breast cancer prognosis. This hyperlipidemic condition may be the reflection of the fat splitting lipases during progression of the disease. In the present study increased lipase activities were observed in mammary carcinoma bearing animals. Elevated activity of lipase was also observed in the malignancies of breast. The increased activities of lipase in patients with breast cancer are associated with increased amounts of subcutaneous body fat as compared with control subjects [41]. Inhibition of lipoprotein lipase activity in cancerous animal is due to high-density lipoprotein, competing with the substrate for binding to the enzyme [42]. Lecithin: cholesterol-acyl transferase (LCAT) mediates the esterification of free cholesterol on plasma lipoproteins. This step converts discoidal, nascent high-density lipoprotein particles into mature spherical high-density lipoprotein. The decreased activity of LCAT maintains low ester cholesterol level in plasma by reducing the efflux of free cholesterol from the proliferating cells and the influx of cholesteryl ester into the cells. The inhibition may indirectly aid the rapid multiplication of the already proliferating cells.

In particular, high levels of triglycerides and free fatty acids were associated with altered levels and distribution of plasma cholesterol, with increased total, very low- and low-density lipoproteins cholesterol and reduced high-density lipoprotein (HDL) cholesterol, which is evident from the current results. Elevated activities of cholesterol ester synthase and cholesterol ester hydrolase, the enzymes that balance the cholesterol levels in the cell and the liberation of cholesterol from lipopro-

tein cholesteryl ester are increased in cancerous animal. The imbalance in the level of cholesterol might be due to the hyperlipidemic condition. The existence of a causal relationship between a high fat content and increased cancer risk has been reported as one of the mechanisms of cellular aberration by interaction with signal transduction pathways, leading to altered gene expression, effects on unbound oestrogenic hormone concentrations (in the case of breast cancer) and structural and functional changes in cell membranes resulting in alterations in hormone and growth factor receptors. In the present study, on treatment with herbal preparation, the reversal trend in the activities of the lipid-metabolizing enzymes was observed. This finding proves that the herbal preparation has hypolipidemic and hypocholesterolemic effect and this effect works on the immune cell and reverses the altered enzyme activities [43].

The low-density lipoprotein –C is more susceptible to oxidation in various pathologic conditions resulting in higher LPO during oxidative stress [44]. The high-density lipoprotein –C, on the other hand, has a great potential to counter the oxidative damage of low-density lipoprotein –C in cell membrane and thereby preventing LPO. It is suggested that high-density lipoprotein –C prevents both enzymatic and nonenzymatic generation of $O_2^{\bullet-}$, H_2O_2 and $\bullet OH$ [45] and thus acts as an anticarcinogen and a powerful antioxidant. In the present study, increased plasma low-density lipoprotein –C and decreased high-density lipoprotein –C levels were observed among the mammary carcinoma animals as compared to the controls. The exact mechanism of hypertriglyceridemia and decreased high-density lipoprotein –C concentration in breast cancer patients is not known. However, it has been suggested that lipoprotein lipase (LPL) may regulate the clearance of triglycerides from blood to tissue and its activity in white adipose tissue decreases in cancer hosts, contributing to the hypertriglyceridemia [46].

Hypertriglyceridemia the common condition in cancerous state [47] is further deranged in cancerous animals. Hypertriglyceridemia is frequently observed in various degrees in tumour bearing animals in combination with increased very low-density lipoprotein and decreased high-density lipoprotein [37]. Defects in the catabolism of triacylglycerol rich lipoproteins by lipoprotein lipase may give rise to hypertriglyceridemia [48]. Significantly higher levels of triglycerides were observed in breast cancer patients when compared with their controls [49]. The data obtained in the present study shows that the lipid levels were reverted back to the near normal levels upon treatment with KA as well as SA. This pharmacological property of the herbal prepara-

tion may be due to the hypolipidemic property and the cytoprotective effect of the drug on the deteriorated cell membrane, which is a crucial condition in the cancerous condition. This property of the herbal preparation may be due to the presence of flavonoids, which have been reported to reduce the levels of serum triglycerides [50].

Plant flavonoids are diphenylpropane derivatives that exert a wide range of biochemical and pharmacological effects. A number of flavonoids have been shown to suppress carcinogenesis in various animal models [51]. There is currently considerable interest in these compounds, as they appear to exert a beneficial effect on several key mechanisms involved in the pathogenesis of cancer. A variety of polyphenolic compounds of dietary origin are known to inhibit cancer. Isoflavonoids are members of the broad class of plant polyphenols that have been shown in vivo to have benefit in the treatment of a wide variety of chronic diseases, including cancer [52]. Another important consideration in vivo is the localized metabolism of isoflavones that may occur in the vicinity of tumour cells. The demonstration that human PMNs stimulated to have a respiratory burst can produce halogenated isoflavones strongly suggests that this may occur in vivo. The consequence is that these locally formed metabolites may have unique biochemical properties and may account for the anti-cancer effects of isoflavones.

KA is found to be more effective than SA in reverting back the altered lipid profile as well as the lipid-metabolizing enzymes. This increased effect is due to the synergistic activities of various components of KA such as EO and honey other than SA.

Ascorbic acid can also protect the deranged lipid and lipoprotein metabolism against oxidative damage. In addition, ascorbic acid may reduce carcinogenesis through the stimulation of immune systems, where cytotoxic T lymphocytes, macrophages and natural killer cells can lyse tumour cells [53]. Plasma levels of ascorbate were significantly lower while platelet levels were higher in a group of recently diagnosed breast cancer patients when compared to a matched group of controls [54]. Epidemiological studies appear to point to ascorbate as a possible chemopreventive for breast cancer. Drug control animals do not show any significant changes, which explain that the herbal preparation does not exert any adverse or deleterious side effects on the cell membranes.

All the above results obtained from the current investigations shows that these herbal preparations KA as well as SA possess a strong hypolipidemic property. They recouped back the deranged lipid metabolism found in cancer condition to normal. The results also shows that

KA is found to be more effective than SA, which may be due to the amalgamated and additive effects of SA, *Emblca officinalis* and honey.

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